

Problem Statement:

You are working with the National Statistical Office which is equipped to release inflation numbers in India. As an analyst, you are provided with CPI data and are equipped to find out insights from the data. Your senior wants you to find key trends and deep dive into the data to answer the following questions -

- 1. Based on the latest month's data, identify the contribution of different broader categories (food, energy, transportation, education, etc.) towards the CPI basket. Broader categories (buckets) can be created by combining similar categories into one bucket; Ex.: Meals, Beverages, Cereals, can be clubbed to create "Food" category, etc.**
 - **Which broader category has the highest contribution towards CPI calculation**
 - **Contribution is calculated by evaluating the underlying index values for broader category and should add to 100% when contribution from different broader categories are added .**
- 2. A trend of Y-o-Y increase in CPI (rural + urban) inflation starting 2017 for the entire basket of products combined.**
 - **Create a graph depicting the growth rate Y-o-Y and identify the year with highest inflation rate**
 - **Highlight the reason why the year has the highest inflation (based on research).**
- 3. With India's retail inflation reaching a 3-month high of 5.55% in November 2023, largely due to a sharp rise in food prices. Analyze the following for 12 months ending May'23**

- ***Investigate trends in the prices of broader food bucket category and evaluate month-on-month changes. Highlight month with highest and lowest food inflation***
 - ***Identify the absolute changes in inflation over the same 12 months period and identify the biggest individual category contributor (only within broader food category) towards inflation***
- 4. *Investigate how the onset and progression of the COVID-19 pandemic affected inflation rates in India. Analyze the Impact of key pandemic milestone (first lockdown) on the CPI inflation %, specially focus on categories like healthcare, food, and essential services.Hint: You can consider Mar'20 as the onset of covid, and can compare the inflation trend before and after Mar'20 to see if there is a change in inflation % before and after.***
- 5. *Investigate how major global economic events (like imported crude oil price fluctuations) have influenced India's inflation. This can include an analysis of imported goods and their price trends.***
- ***For the purpose of this analysis, focus only on the imported oil price fluctuations for years 2021 to 2023 (Month-on-month)***
 - ***Identify trends in oil price change with change in inflation prices of all the categories and identify category whose inflation prices strongly changes with fluctuations in imported oil price (Hint: you can use =correl function)***

Other Information:

- ***Percentage Change: To understand inflation or deflation trends, calculate the percentage change in CPI between two periods***

(e.g., year-over-year or month-over-month). This shows how much prices have increased or decreased relative to the earlier period.

- ***For example, to calculate the monthly inflation rate between two consecutive months:***
 - ***Monthly Inflation rate = $\frac{\text{CPI in current month} - \text{CPI in previous month}}{\text{CPI in previous month}} \times 100$***
- ***Annual Inflation Rate: For longer periods, such as yearly inflation, use the CPI values at the start and end of the period. This helps in understanding the overall inflation experienced over the year.***
 - ***Annual Inflation rate = $\frac{(\text{CPI at end of year} - \text{CPI at start of year})}{\text{CPI at start of year}} \times 100$***
 - ***Any month can be considered as start of the year, then end of the year month will be considered 12 months after the month you have selected for start of the year***